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标题	Design of Experimental Conditions with Machine Learning for Collaborative Organic Synthesis Reactions Using Transition-Metal Catalysts
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摘要（原文节选）

To improve product yields in synthetic reactions, it is important to use appropriate catalysts. In this study, we used machine learning to design catalysts for a reaction system in which both Buchwald – Hartwig-type and Suzuki – Miyaura-type cross-coupling reactions proceed simultaneously. First, using an existing dataset, yield prediction models were constructed with machine learning between experimental conditions, including the substrate and catalyst and the yields of the two products. Seven methods for calculating ⋯

数据表 · Table 1（7 行，含数值行 6 行）

Calculated rDCV2 and MAEDCV of the Best Models for Each Yield Using Our Dataset

	4 (BHCC)	5 (SMCC)				
	regression method	rDCV2	MAEDCV	regression method	rDCV2	MAEDCV
Method A	XGB	0.449	11.0	XGB	0.508	15.5
Method B	XGB	0.099	15.4	GP	0.276	19.6
Method C	XGB	0.783	7.4	LASSO	0.563	15.2
Method D	RF	0.713	8.5	LASSO	0.495	16.4
Method E	RF	0.574	10.2	RF	0.507	16.2
Method F	LASSO	0.530	11.6	EN	0.446	16.1

正文对应句（逐字）：Our dataset consisted of 73 experiments with different experimental conditions for the substrates and catalysts. The y-variables are the yields [%] of compound 4 (BHCC) and compound 5 (SMCC), as shown in Figure 1. Since the yield of compound 4 did not exceed 10% when X was Cl (aryl chloride) in subs

数据表 · Table 2（4 行，含数值行 4 行）

Predicted and Experimental Yields for New Combinations of Substrates and Catalysts

		sample 1a	sample 2b
yield of 4 (BHCC) [%]	prediction	70	0
	experiment	71	0
yield of 5 (SMCC) [%]	prediction	14	94
	experiment	16	96

脚注（逐字）：aR is OMe, X is Cl, and the catalyst is Ni(acac)2(tolNH2)2.bR is Me, X is I, and the catalyst is Pd(dppf)Cl2.

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